

STEP 3 — OMNICHANNEL CONVERSATION SYSTEM (GHL Conversations Clone)

This is what clients LOVE in GHL:

“All messages (WhatsApp, Email, SMS) in ONE inbox”

STEP 3 GOAL

Build a **unified inbox** where:

- All conversations are stored
 - Real-time chat works
 - Messages sync with CRM
 - Automation can read replies
-

3.1 ARCHITECTURE OVERVIEW

User (WhatsApp / Email / SMS)
↓
Provider API (Meta / SMTP / Twilio)
↓
Webhook → Your Server (Perfex)
↓
Save in `tbl_conversations`
↓
Realtime UI (Inbox)
↓
Automation (n8n reacts)

3.2 DATABASE FINAL STRUCTURE

You already created `tbl_conversations`, now upgrade it:

✓ Updated Table

```
ALTER TABLE tbl_conversations
ADD message_id VARCHAR(191) NULL,
ADD sender VARCHAR(191) NULL,
ADD receiver VARCHAR(191) NULL,
ADD is_read TINYINT(1) DEFAULT 0,
ADD media_url TEXT NULL,
ADD message_type VARCHAR(50) DEFAULT 'text';
```

✓ Final Structure

tbl_conversations

- id
- lead_id
- message
- direction (inbound/outbound)
- channel (whatsapp/email/sms)
- sender
- receiver
- message_id
- message_type
- media_url
- status
- is_read
- created_at

➡ 📱 3.3 WHATSAPP INTEGRATION (META CLOUD API)

🔥 Why Meta API?

- Cheap (almost free)
 - Direct WhatsApp integration
 - Scalable
-

✓ Setup Steps

1. Go to: <https://developers.facebook.com>
 2. Create App
 3. Add **WhatsApp Cloud API**
 4. Get:
 - Access Token
 - Phone Number ID
 - Webhook URL
-

3.4 WEBHOOK (RECEIVE MESSAGES)

✓ Endpoint in Perfex

```
public function whatsapp_webhook()
{
    $input = json_decode(file_get_contents('php://input'), true);

    $message = $input['entry'][0]['changes'][0]['value']['messages'][0] ?? null;

    if ($message) {
        $phone = $message['from'];
        $text = $message['text']['body'] ?? '';

        // Find lead by phone
        $lead = $this->db->where('phonenumber', $phone)->get('tblleads')->row();

        if ($lead) {
            // Save message
            $this->db->insert('tbl_conversations', [
                'lead_id' => $lead->id,
                'message' => $text,
                'direction' => 'inbound',
                'channel' => 'whatsapp',
                'sender' => $phone,
                'receiver' => 'business',
                'message_id' => $message['id']
            ]);

            // Trigger automation
```

```

        trigger_event('message_received', [
            'lead_id' => $lead->id,
            'message' => $text
        ]);
    }
}

echo "OK";
}

```



3.5 SEND WHATSAPP MESSAGE

✓ API Function

```

function send_whatsapp_message($to, $message)
{
    $token = "YOUR_ACCESS_TOKEN";
    $phone_number_id = "YOUR_PHONE_ID";

    $url = "https://graph.facebook.com/v17.0/$phone_number_id/messages";

    $data = [
        "messaging_product" => "whatsapp",
        "to" => $to,
        "type" => "text",
        "text" => ["body" => $message]
    ];

    $ch = curl_init($url);
    curl_setopt($ch, CURLOPT_HTTPHEADER, [
        "Authorization: Bearer $token",
        "Content-Type: application/json"
    ]);
    curl_setopt($ch, CURLOPT_POSTFIELDS, json_encode($data));
    curl_exec($ch);
}

```

✓ Save Outgoing Message

```

$this->db->insert('tbl_conversations', [
    'lead_id' => $lead_id,
    'message' => $message,

```

```
'direction' => 'outbound',  
'channel' => 'whatsapp'  
]);
```

3.6 EMAIL INTEGRATION (2-WAY)

 **Sending → already done (Step 2)**

 **Receiving Emails (IMPORTANT)**

Use:

- **IMAP (Gmail / SMTP inbox reading)**
-

Example:

```
$inbox = imap_open('{imap.gmail.com:993/imap/ssl}INBOX', $email, $password);
```

```
$emails = imap_search($inbox, 'UNSEEN');
```

```
foreach ($emails as $mail) {  
    $overview = imap_fetch_overview($inbox, $mail, 0);  
    $message = imap_fetchbody($inbox, $mail, 1);
```

```
    // Match lead by email  
    // Save in tbl_conversations  
}
```

3.7 UNIFIED INBOX UI (IMPORTANT)

 **Page:**

/admin/inbox



Layout:

Leads List Chat Window	
----- -----	
John Hello	
Mike Hi there	



Left Panel (Leads)

Query:

```
SELECT lead_id, MAX(created_at) as last_msg
FROM tbl_conversations
GROUP BY lead_id
ORDER BY last_msg DESC
```



Right Panel (Messages)

```
$this->db->where('lead_id', $lead_id);
$this->db->order_by('created_at', 'ASC');
$messages = $this->db->get('tbl_conversations')->result();
```



3.8 REAL-TIME CHAT (IMPORTANT)

Option 1: Pusher (easy)

Option 2: Socket.io (advanced)



Pusher Example

When message inserted:

```
$pusher->trigger('chat-channel', 'new-message', [
    'lead_id' => $lead_id,
    'message' => $message
]);
```

JS:

```
channel.bind('new-message', function(data) {  
  if (data.lead_id == currentLead) {  
    appendMessage(data.message);  
  }  
});
```



3.9 READ / UNREAD SYSTEM



Update on open:

```
$this->db->where('lead_id', $lead_id);  
$this->db->update('tbl_conversations', ['is_read' => 1]);
```



3.10 SEARCH + FILTER

Add filters:

- By channel (WhatsApp / Email)
 - By unread
 - By staff assigned
-



3.11 AUTOMATION INTEGRATION

When message received:

```
trigger_event('message_received', [  
  'lead_id' => $lead_id,  
  'message' => $text  
]);
```

Example Automation:

- If message contains "price" → send pricing
 - If no reply → follow-up
-



3.12 TESTING CHECKLIST



Must Work:

- WhatsApp incoming message saved
 - WhatsApp outgoing works
 - Email fetch working
 - Inbox UI loads
 - Real-time updates working
 - Read/unread works
 - Automation triggers on reply
-



STEP 3 OUTPUT

After this:



GHL-style Conversations Inbox



WhatsApp integration



Email integration



Real-time chat



Automation-ready messaging